

Mock Data — Schema & Demo Scenario

The demo scenario: "Typhoon Yagi has closed the Port of Tanjung Pelepas (Malaysia) for 72 hours. Primary supplier SUP-001 (Chang Electronics, Malaysia) cannot fulfil 23 active orders totalling 48,200 units of automotive-grade MCUs and sensor components."

3.1 M1: orders collection (20–30 documents)

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Purpose	Source for blast_radius_query \$graphLookup. Must contain realistic BOMs with depends_on graph edges pointing to supplier_ids that will be disrupted in the demo scenario.	
Field	Type / value	Notes
_id	ObjectId	Auto-generated
order_number	String	"ORD-1044" through "ORD-1073"
status	String	"active" — all seed orders must be active to appear in blast radius
sku_ids	Array<String>	["MCU-32X", "SEN-IMU9", "CAP-470UF"] — 3–4 distinct SKUs
supplier_id	String	Must match _id of a disrupted supplier in suppliers collection — this is the graph edge
quantity_ordered	Number	5000–20000 per order
quantity_remaining	Number	80–95% of quantity_ordered — simulates partially fulfilled orders
route	Object	{ origin: "Shanghai", destination: "Hamburg", via_port: "Singapore" }
bom	Object	Polymorphic — microchip BOM: { part_number, spec, grade }
depends_on	Array<String>	[supplier_id] — the graph edge field used by \$graphLookup
unit_price	Number	Original unit price from disrupted supplier — used for cost variance calc

created_at	Date	ISODate within last 90 days
Demo note	Ensure 15–20 orders have depends_on pointing to the "disrupted supplier" (SUP-001 in demo scenario). Remaining 10 orders point to other suppliers to show selective blast radius.	

3.2 M2: suppliers collection (15–20 documents)

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Purpose	Source for supplier_vector_search \$vectorSearch. Must include: 1 disrupted supplier (SUP-001), 1 supplier that fails ESG (SUP-FAIL), and 2–3 compliant backup suppliers with sufficient capacity.	
Field	Type / value	Notes
_id	String	"SUP-001" (disrupted), "SUP-002" through "SUP-020"
name	String	Realistic company names e.g. "PT Maju Jaya Electronics"
capabilities	Array<String>	["automotive-grade microcontrollers", "32-bit MCU", "AEC-Q100 certified"]
capacity_units_month	Number	Must be >= required_capacity (48200) for at least 2 backup suppliers
geo_region	String	"Indonesia", "Malaysia", "South Korea" — avoid disrupted region (SE Asia via_port)
lead_time_days	Number	14–45 days
esg_score	Number	SUP-FAIL: 22 (triggers compliance fail). Others: 65–92
certifications	Array<String>	["ISO-14001", "carbon-neutral-2024"] for compliant suppliers
unit_price	Number	5–15% higher than original supplier — realistic cost variance
embedding	Array<Float>	Auto-generated by Atlas Voyage AI provider on insert — do not populate manually

last_updated	Date	ISODate — current date
Demo note	SUP-001 is the "disrupted" supplier — its _id appears in impacted orders. SUP-FAIL has esg_score=22 and geo_region="Vietnam" to trigger the ESG compliance retry in the demo. SUP-002 ("PT Maju Jaya") is the winning compliant supplier.	

3.3 M3: compliance_policies collection (30–50 chunks)

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Purpose	Source for compliance_rag_query \$vectorSearch RAG. Must contain chunks that will (a) flag SUP-FAIL on ESG/labour, and (b) pass SUP-002 on all policy dimensions. Generated by running the PDF preprocessing script on sample policy documents.	
Field	Type / value	Notes
_id	ObjectId	Auto-generated
source_doc	String	"2025_ESG_Guidelines.pdf", "Trade_Sanctions_2025.pdf", "Labour_Standards.pdf"
chunk_index	Number	0-based index within source document
text	String	300–500 token chunk of policy text — must contain explicit Vietnam labour risk language for demo
policy_type	String	"ESG" "trade_sanction" "labour" "carbon"
embedding	Array<Float>	Auto-generated by Atlas Voyage AI provider on insert
effective_date	Date	2025-01-01
Demo note	Create at least 3 chunks containing text about "Vietnam child labour risk" or "Vietnam manufacturing ESG concerns" — these will be retrieved when compliance_rag_query runs for SUP-FAIL (geo_region: Vietnam). Create 3 chunks confirming Indonesia and South Korea as compliant regions.	